

COMS 4030A and COMS 7047A
Adaptive Computation and Machine Learning

TEST 11 April 2024

Time: One Hour

Question 1. (4 marks)

Let w_{mn} denote the weight of the edge from node m to node n in a neural network. Using the notation from the Lecture Notes, explain how the gradient descent update rule for the current weight $\underline{w}_{mn}$ simplifies to the following:

$$\underline{w}_{mn} \leftarrow \underline{w}_{mn} - \eta \delta_n a_m.$$

Question 2. (4 marks)

Consider the cross-entropy loss function, where ℓ ranges over all output nodes,

$$L_{CE}(\mathbf{t}, \mathbf{y}) = - \sum_{\ell} t_{\ell} \ln(y_{\ell}).$$

Suppose y_{ℓ} is obtained by *softmax* activation as follows, where m ranges over all output nodes:

$$y_{\ell} = \text{softmax}(z_{\ell}) = \frac{e^{z_{\ell}}}{\sum_m e^{z_m}}.$$

(a) (2 marks) Show that L_{CE} simplifies to

$$L_{CE}(\mathbf{t}, \mathbf{y}) = - \sum_{\ell} t_{\ell} z_{\ell} + \ln \sum_m e^{z_m}.$$

(b) (2 marks) Use your answer to (a) to show that for a node n ,

$$\frac{\partial L_{CE}}{\partial z_n} = -t_n + y_n.$$

Question 3. (2 marks)

Consider the following confusion matrix obtained by applying a model to a binary classification dataset with classes 1 and 0 denoting positive and negative, respectively:

		prediction	
		1	0
actual	1	233	21
	0	42	187

Compute the following values with respect to the above confusion matrix:

(i) accuracy, (ii) precision, (iii) sensitivity and (iv) F1 score.

Question 4. (4 marks)

Explain what L_2 -regularization is and how it is used.

Question 5. (4 marks)

Use the Entropy method to create a decision tree with the dataset S given below.

Show all your calculations and draw the final decision tree.

	F_1	F_2	target
	a	0	Y
	c	1	N
	c	1	Y
	b	0	N
	a	0	Y
$S :$	b	1	N
	b	1	Y
	c	0	Y
	c	1	N
	a	1	N
	c	0	Y
	b	0	N

Question 6. (3 marks)

Compute $\text{Gain}(T, F_1)$ for the dataset T below, using the mean square error and split point of $\frac{2}{3}$ for attribute F_1 . Show all your calculations.

	F_1	target
	0.9	0.8
	0.6	0.4
$T :$	0.5	0.7
	0.3	0.6
	0.7	0.5
	0.8	0.7

Question 7. (3 marks)

(a) (1 marks) What is the MAP hypothesis?

(b) (2 marks) Use the naïve Bayes classifier on the following dataset to classify input $(1, b, F)$.

Show your working.

F_1	F_2	F_3	target
0	a	T	Y
1	b	T	N
1	c	F	N
1	c	F	Y
0	b	T	N
0	a	F	N
1	a	T	Y
0	b	F	Y
1	c	T	N
0	c	F	Y
1	b	T	Y

Total marks: 24